

AP CALCULUS AB – UNIT 5

5.1 Mean Value Theorem



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Verify continuity and differentiability hypotheses for the Mean Value Theorem.
- Find or guarantee points where instantaneous and average rates agree.
- Apply Rolle's Theorem as a special case.
- Distinguish existence conclusions from exact location and from the Intermediate Value Theorem.

Keys: Essential Knowledge and Vocabulary

Mean Value Theorem	If f is continuous on $[a, b]$ and differentiable on (a, b) , then some $c \in (a, b)$ satisfies $f'(c) = \frac{f(b)-f(a)}{b-a}$.
Rolle's Theorem	If the MVT hypotheses hold and $f(a) = f(b)$, then some $c \in (a, b)$ has $f'(c) = 0$.
Geometric meaning	At some interior point, a tangent line is parallel to the secant through the endpoints.
Existence statement	The theorem guarantees at least one c but does not necessarily identify it or guarantee uniqueness.
Hypothesis failure	A corner, cusp, discontinuity, or vertical tangent can prevent use of the theorem.
IVT distinction	IVT concerns attained function values; MVT concerns derivative values matching an average slope.

Explanation, Strategy, and Common Errors

- State both hypotheses on their correct intervals before writing the conclusion.
- Compute the endpoint secant slope, then solve $f'(c)$ equal to that slope and keep only solutions in (a, b) .
- Do not use endpoint values alone to claim differentiability.
- A theorem may be inapplicable even if a point satisfying the conclusion happens to exist.
- In context, interpret the average and instantaneous rates with units.

Solved Example: Polynomial MVT

Verify the MVT for $f(x) = x^2 - 2x$ on $[0, 3]$ and find c .

Answer and Reasoning

A polynomial is continuous and differentiable. The secant slope is $(3 - 0)/3 = 1$. Since $f'(x) = 2x - 2$, solve $2c - 2 = 1$ to get $c = 3/2$.

Solved Example: Rolle's Theorem

Apply Rolle's Theorem to $f(x) = x^3 - 4x$ on $[-2, 2]$.

Answer and Reasoning

The polynomial is continuous/differentiable and $f(-2) = f(2) = 0$. Solve $f'(c) = 3c^2 - 4 = 0$:
 $c = \pm 2/\sqrt{3}$, both interior.

Solved Example: Hypothesis failure

Can the MVT be applied to $f(x) = |x|$ on $[-1, 1]$?

Answer and Reasoning

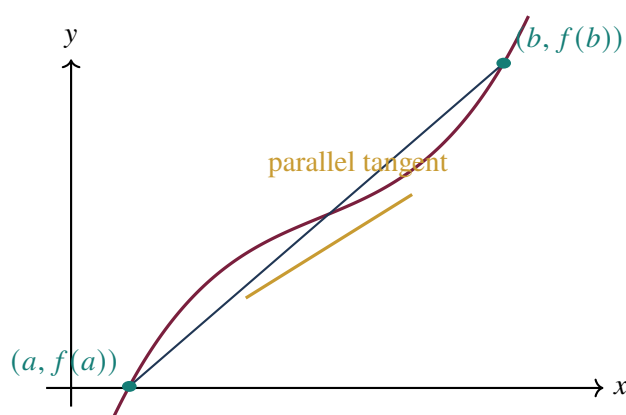
No. The function is continuous, but it is not differentiable at $x = 0$.

Solved Example: Context

A car travels 180 km in 2 hours with differentiable position. What does the MVT guarantee?

Answer and Reasoning

At some time during the trip, instantaneous velocity equals the average velocity 90 km/h.

Shared Representation / Data**AP Exam Language**

For theorem applications, state every hypothesis. For graph analysis, justify sign changes and compare all required candidates. For optimization, state the objective, constraint, feasible domain, and verification.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. State the two MVT hypotheses.

2. Find the average slope of $f(x) = x^2$ on $[1, 4]$.

3. For the previous function, solve $f'(c) = 5$.

4. State Rolle's conclusion when its hypotheses hold.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Apply the MVT to $f(x) = 1/x$ on $[1, 4]$ and find c .

6. Determine whether MVT applies to $f(x) = \sqrt{x}$ on $[0, 4]$.

7. Does MVT apply to $f(x) = 1/(x - 2)$ on $[1, 3]$?

8. A differentiable temperature changes from 18°C to 30°C in 4 hours. State the guarantee.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Find all MVT points for $f(x) = x^3 - 3x$ on $[-2, 2]$.

10. Explain why MVT does not guarantee exactly one point c .

11. If $f'(x) = 0$ for all x in an interval, prove f is constant there using MVT.

12. If $|f'(x)| \leq 4$ on $[1, 6]$, bound $|f(6) - f(1)|$.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Prove $x^5 + x + 1 = 0$ has at most one real root.

14. Suppose $f(0) = 2$ and $|f'| \leq 3$. Give the tight MVT interval containing $f(0.4)$.

15. A function is continuous on $[a, b]$ and differentiable except at one interior corner. Can MVT still be invoked?

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Use MVT to prove $|\sin b - \sin a| \leq |b - a|$.

17. If $f'(x) > g'(x)$ for all x in (a, b) and $f(a) = g(a)$, prove $f(x) > g(x)$ for $x > a$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. MVT requires a function to be

- | | |
|---|--------------------------------|
| (A) continuous on (a, b) only | (B) differentiable on $[a, b]$ |
| (C) continuous on $[a, b]$ and differentiable on (a, b) | (D) increasing on $[a, b]$ |

2. Rolle's Theorem is the MVT case in which

- | | |
|---------------------|-------------------|
| (A) $f'(a) = f'(b)$ | (B) $f(a) = f(b)$ |
| (C) $a = b$ | (D) f is linear |

3. For $f(x) = x^2$ on $[0, 2]$, the MVT point is

- | | |
|----------------|-------|
| (A) 0 | (B) 1 |
| (C) $\sqrt{2}$ | (D) 2 |

4. The theorem guarantees

- | | |
|-----------------------|---------------------------------|
| (A) exactly one point | (B) at least one interior point |
| (C) an endpoint | (D) a maximum |

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. For $f(x) = \ln x$ on $[1, 5]$, the MVT point is closest to

- | | |
|----------|----------|
| (A) 1.61 | (B) 2.49 |
| (C) 3.11 | (D) 4.00 |

6. A position changes by 126.4 km over 1.6 h. MVT guarantees velocity equals

- | | |
|-----------|-----------|
| (A) 63 | (B) 79 |
| (C) 101.1 | (D) 202.2 |

AP-Style Free-Response Practice

No Calculator

Let $f(x) = x^3 - 3x + 2$ on $[-2, 1]$.

(a) Verify the MVT hypotheses. [2 points]

(b) Find all c satisfying the MVT conclusion. [3 points]

(c) Explain whether Rolle's Theorem also applies. [2 points]

Calculator Permitted

A differentiable vehicle position $s(t)$ satisfies $s(0) = 12$ km and $s(2.5) = 207$ km.

(a) Find average velocity. [2 points]

(b) State the MVT conclusion in context. [2 points]

(c) If the speed limit was 70 km/h throughout, explain the implication. [2 points]